Simulation of Squeezed Light Generation in an Optical Resonator

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Why?
Quantum Internet

There are existing fiber optic networks all over the world which can transport optically encoded information at the speed of light.

Current methods for encoding quantum information in optical beams produce photons at wavelengths not compatible with current infrastructure.

Reduce

To rapidly deploy a quantum internet across Canada and the world, a cost effective and reliable method for converting optically-encoded continuous-variable quantum information to telecom-compatible wavelengths is required[1].

Fig. 1 – Existing Bell Fiber Optic Network [2].

A potential candidate is hot rubidium-87 gas which may be able to convert information encoded in a 795 nm beam (which can be produced with modern techniques) to a 1530 nm beam (which is within the window of the telecom
C-band).

The wavelength-conversion happens via optical ***Four Wave Mixing*** and can be resonantly enhanced via ***Cavity Electromagnetically Induced Transparency*** at the telecom wavelength.

Four Wave Mixing involves driving an atomic sample by multiple beams to induce the emission of a separate beam of ***Squeezed Light*** such that the conservation equation and momentum conservation equations are satisfied.

$$\omega_s + \omega_{II} = \omega_t + \omega_I$$
$$k_s + k_t + k_I + k_{II} = 0$$

X^(3)

Cavity Electromagnetically Induced Transparency involves driving the atomic sample within an enhancing resonant cavity such that beams interfere to produce a polarization field which allows one frequency of light to pass through the medium with minimal losses.

How?
Setup and Simulation Methods

We must computationally model the quantum transitions of a hot gas composed of 4-level Diamond-Configuration atoms driven by multiple lasers tuned near-resonance of atomic transitions: two pump lasers stimulating transitions and one signal laser encoded with quantum information.

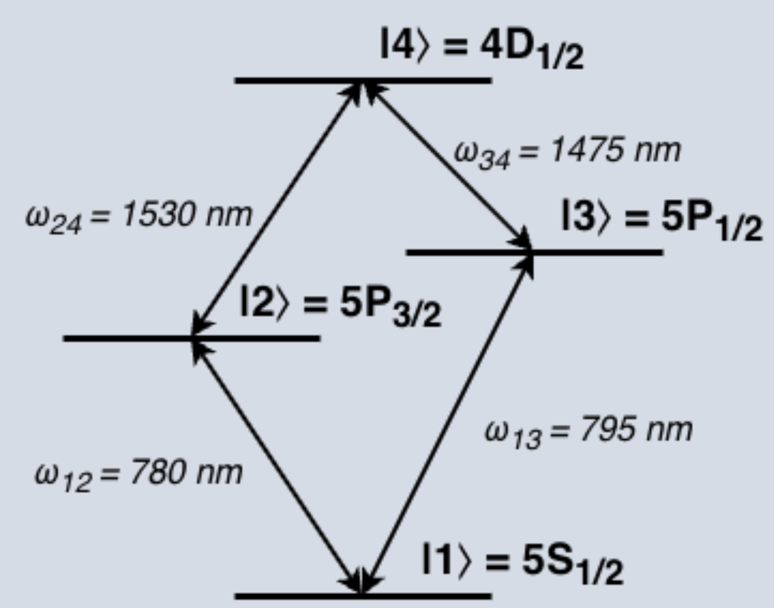


Fig. 2 – Diamond-Configuration Atomic Transition Structure.

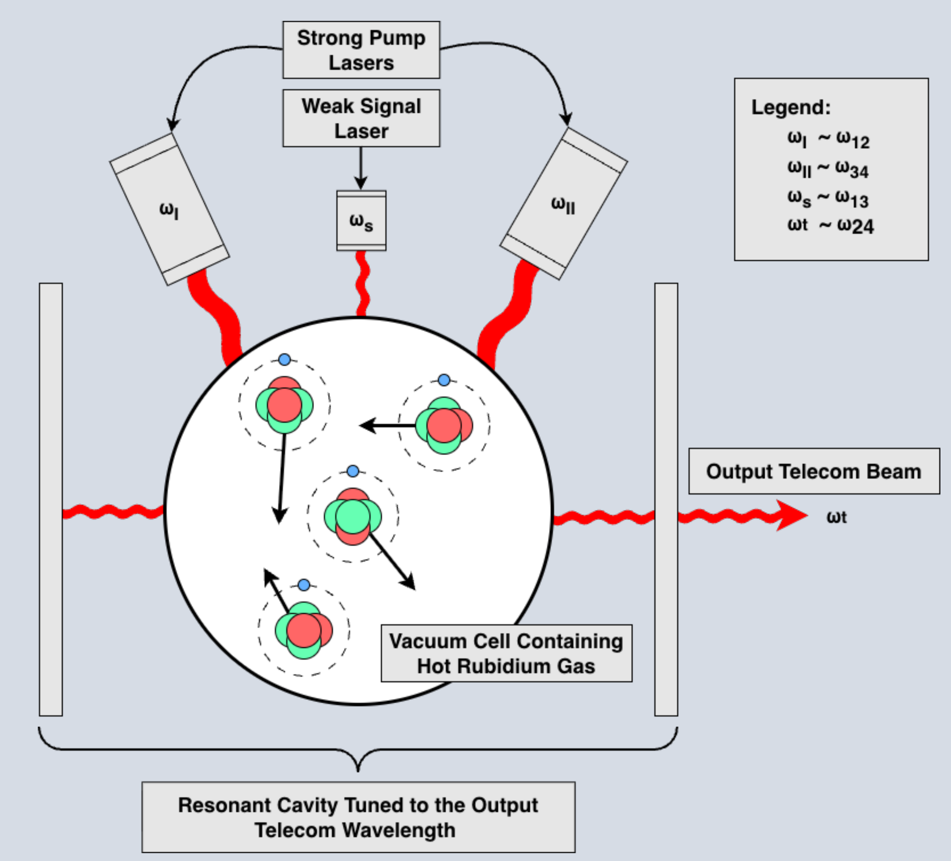


Fig. 3 – Vacuum Chamber Driven within a Resonant Cavity Driven by Lasers Tuned near Transition Frequencies.

The overarching goal is to find system parameterizations which induce nonlinear optical processes to create quantum optical correlations known as ***Squeezed Light***, which is light with reduced uncertainty in one quadrature (amplitude or phase) but increased uncertainty in the other such that the ***Generalized Uncertainty Principle*** is still observed.

Picture of Squeezing

$$\Delta\text{Amplitude} \cdot \Delta\text{Phase} \geq \frac{\hbar}{2}$$

The objective is to use the following Hamiltonian describing the interaction between the atoms and the external fields, with atomic velocities $\vec{v}_i$ distributed as a ***Multivariate Normal Distribution*** which results in ***Doppler Broadening***.

$$\tilde{H} = \sum_{i=1}^{N} \hat{\mathbb{1}}^{\otimes(i-1)} \otimes \hat{H}^{(i)}_{\text{atom}} \otimes \hat{\mathbb{1}}^{\otimes(N-i)} + \hbar(\omega_{13} - \omega_s)\hat{a}^\dagger\hat{a} + \hbar(\omega_{24} - \omega_t)\hat{b}^\dagger\hat{b}$$

Doppler Broadening in bins

$$\hat{H}^{(i)}_{\text{atom}} = \hbar \begin{pmatrix} 0 & \Omega_I & \Omega_s\hat{a} & 0 \\ \Omega_I^* & (\omega_{12} - \omega_I + \vec{k}_{12}\cdot\vec{v}_i) & 0 & \Omega_t\hat{b} \\ \Omega_s^*\hat{a}^\dagger & 0 & (\omega_{13} - \omega_s + \vec{k}_{13}\cdot\vec{v}_i) & \Omega_{II} \\ 0 & \Omega_t^*\hat{b}^\dagger & \Omega_{II}^* & (\omega_{34} - \omega_{II} + \vec{k}_{34}\cdot\vec{v}_i) + (\omega_{13} - \omega_s + \vec{k}_{13}\cdot\vec{v}_i) \end{pmatrix}$$

This is done by solving the ***Lindblad Master Equation*** (below) for the steady state dynamics (in particular, values of ρ_{13} and ρ_{24}) of the system at various hyper-parameter configurations to find when ***Electromagnetically Induced Transparency*** and ***Four Wave Mixing*** phenomena occur simultaneously to generate ***Squeezed Light***.

$$\frac{d\hat{\rho}}{dt} = -\frac{i}{\hbar}\left[\tilde{H}, \hat{\rho}\right] + \sum_i \gamma_i \left(\hat{L}_i\hat{\rho}\hat{L}_i^\dagger - \frac{1}{2}\left\{\hat{L}_i^\dagger\hat{L}_i, \hat{\rho}\right\}\right)$$

state ρ depends on Ω value

These steady state values are then applied to Maxwell's equations in a co-propagating frame to form the ***Maxwell-Bloch Equations*** yielding strength of the telecom-wavelength field envelope as it propagates through the medium [3].

$$\partial_z E_s = i\frac{\omega_s|\hat{d}_{13}|^2}{2\epsilon_0\hbar}\hat{\rho}_{13}$$

No hat on ρ

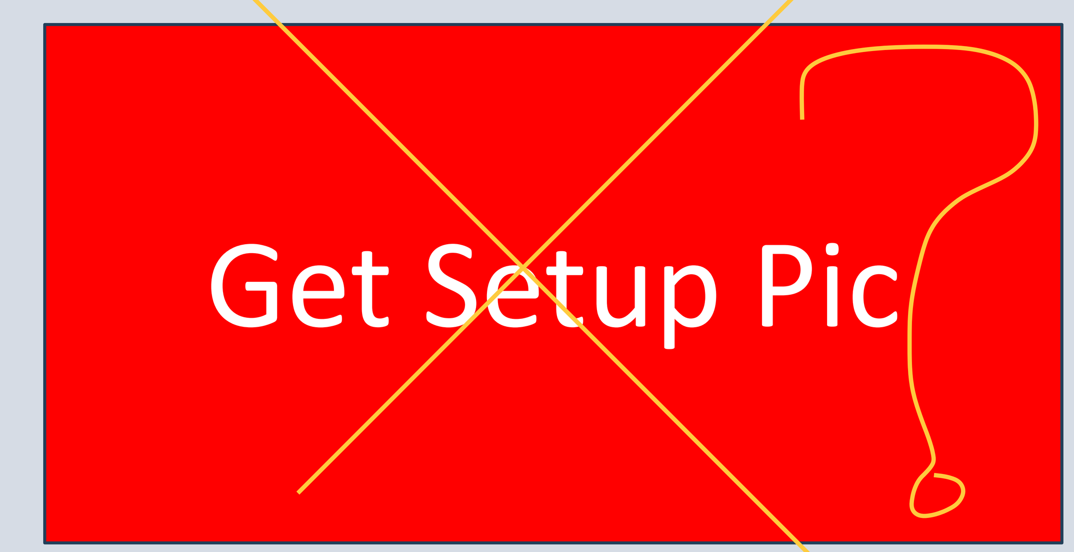
$$\partial_z E_t = i\frac{\omega_t|\hat{d}_{24}|^2}{2\epsilon_0\hbar}\hat{\rho}_{24}$$

Replace E by Ω
Value of Ω depends on ρ

This gives us a theoretical value for the ***Conversion Efficiency*** of the signal beam to the telecom-wavelength transmission beam, and a high ***Squeezing Amplitude*** makes the quantum information less prone to random fluctuations.

Then What?
Takeaways and Outlook

- With a theoretical value of ***Conversion Efficiency*** and ***Squeezing Amplitude*** either experimental research can be justified or the relatively expensive alternative of using a super-cooled atomic system is supported.
- The ***System Parameterizations*** which yield optimal results can be used as a starting point for experiments.
- If hot atomic ensembles can be used to effectively transmit telecom signals, then connecting nodes in a quantum communication network would be fast, economically viable, and compatible with existing techniques.



Legend

ω ≡ Electromagnetic Field Frequency or Transition Frequency
Ω ≡ Rabi Frequency
$\vec{k}$ ≡ Electromagnetic Field Wavevector
$\hat{a}$ ≡ Annihilation Operator of Signal Field
$\hat{b}$ ≡ Annihilation Operator of Telecom-Wavelength Idler Field
$\hat{\rho}$ ≡ Density Matrix
$\hat{L}_i$ ≡ i^{th} Decay Operator
γ_i ≡ Decay Rate of i^{th} Decay Operator
E ≡ Electromagnetic Envelope

References

[1]: Du, D., Castillo-Veneros, L., Cottrill, D., Cui, G., Stankus, P., Katramatos, D., Martínez-Rincón, J., & Figueroa, E. (2021). A long-distance quantum-capable internet testbed. *arXiv (Cornell University)*. https://doi.org/10.48550/arxiv.2101.12742

[2]: Canada's most extensive network coverage. Our Network | Bell Wholesale. (n.d.). https://wholesale.bell.ca/our-network

[3]: MacRae, A., & Kupchak, C. (2026). Propagation dynamics and transient amplification in warm and cold atomic EIT systems. *Optics Communications*, *606*, 132893. https://doi.org/10.1016/j.optcom.2026.132893